

量子力学作业

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第三章

21. 已知波函数:

$$\Psi(x, t) = \sqrt{\frac{2a}{\pi}} \frac{1}{\gamma} \exp \left[-\frac{a}{\gamma^2} \left(x - \frac{hlt}{m} \right)^2 + il \left(x - \frac{hlt}{2m} \right) \right], \quad \gamma = \sqrt{1 + \frac{2ai\hbar t}{m}}$$

注意到 $\left(\frac{z}{w}\right)^* = \frac{z^*}{w^*}$ 和 $(\sqrt{a+ib})^* = \sqrt{a-ib}$, 记 $\theta = \frac{2a\hbar t}{m}$, 则 $\gamma = \sqrt{1+i\theta}$, 有:

$$|\gamma|^2 = \sqrt{(1+i\theta)(1-i\theta)} = \sqrt{1+\theta^2}, \quad \left(\frac{1}{\gamma}\right)^* = \frac{1}{\gamma^*} = \frac{1}{\sqrt{1-i\theta}}$$

可得:

$$\begin{aligned} |\Psi(x, t)|^2 &= \sqrt{\frac{2a}{\pi}} \frac{1}{|\gamma|^2} \exp \left[-2a \operatorname{Re} \left(\frac{1}{\gamma^2} \right) \left(x - \frac{hlt}{m} \right)^2 \right] \\ &= \sqrt{\frac{2a}{\pi}} \frac{1}{|\gamma|^2} \exp \left[-2a \left(\frac{1}{1+\theta^2} \right) \left(x - \frac{hlt}{m} \right)^2 \right] \\ &= \sqrt{\frac{2a}{\pi}} \frac{1}{|\gamma|^2} \exp \left[-\frac{2a}{|\gamma|^4} \left(x - \frac{hlt}{m} \right)^2 \right] \end{aligned}$$

则 x 的平均值为:

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{+\infty} \langle \Psi | x | \Psi \rangle dx \\ &= \int_{-\infty}^{+\infty} x |\Psi|^2 dx \\ &= \int_{-\infty}^{+\infty} \sqrt{\frac{2a}{\pi}} \frac{1}{|\gamma|^2} x \exp \left[-\frac{2a}{|\gamma|^4} \left(x - \frac{hlt}{m} \right)^2 \right] dx \\ &= \int_{-\infty}^{+\infty} \sqrt{\frac{2a}{\pi}} \frac{1}{|\gamma|^2} \left(u + \frac{hlt}{m} \right) e^{-\frac{2a}{|\gamma|^4} u^2} du, \quad u = x - \frac{hlt}{m} \end{aligned}$$

令 $\lambda = \frac{\sqrt{2a}}{|\gamma|^2}$, 由:

$$\int_{-\infty}^{+\infty} e^{-u^2} du = \sqrt{\pi}, \quad \int_{-\infty}^{+\infty} u e^{-u^2} du = 0$$

得:

$$\begin{aligned}
 \langle x \rangle &= \int_{-\infty}^{+\infty} \frac{\lambda}{\sqrt{\pi}} \left(u + \frac{hlt}{m} \right) e^{-\lambda^2 u^2} du \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} \lambda u e^{-\lambda^2 u^2} du + \frac{hlt}{m} \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} \lambda e^{-\lambda^2 u^2} du \\
 &= \frac{1}{\lambda \sqrt{\pi}} \int_{-\infty}^{+\infty} \lambda u e^{-\lambda^2 u^2} d\lambda u + \frac{hlt}{m} \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\lambda^2 u^2} d\lambda u \\
 &= 0 + \frac{hlt}{m} \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \\
 &= \frac{hlt}{m}
 \end{aligned}$$

根据 Ehrenfest 定理有:

$$\frac{d\langle x \rangle}{dt} = \frac{hl}{m}, \quad p = m \frac{d\langle x \rangle}{dt} = hl$$

同理, 由:

$$\int_{-\infty}^{+\infty} u^2 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

得 x^2 的平均值:

$$\begin{aligned}
 \langle x^2 \rangle &= \int_{-\infty}^{+\infty} \langle \Psi | x^2 | \Psi \rangle dx \\
 &= \int_{-\infty}^{+\infty} \frac{\lambda}{\sqrt{\pi}} \left(u + \frac{hlt}{m} \right)^2 e^{-\lambda^2 u^2} du \\
 &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} \lambda u^2 e^{-\lambda^2 u^2} du + \left(\frac{hlt}{m} \right) \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} 2\lambda u e^{-\lambda^2 u^2} du + \left(\frac{hlt}{m} \right)^2 \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} \lambda e^{-\lambda^2 u^2} du \\
 &= \frac{1}{\lambda^2 \sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} + 0 + \left(\frac{hlt}{m} \right)^2 \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \\
 &= \frac{1}{2\lambda^2} + \left(\frac{hlt}{m} \right)^2
 \end{aligned}$$

于是有:

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda^2} = \frac{1}{4a} + \frac{a\hbar^2 t^2}{m^2}$$

对于自由粒子, $H = \frac{p^2}{2m}$, 所以:

$$\langle H \rangle = \int_{-\infty}^{+\infty} \langle \Psi | x^2 | \Psi \rangle dx = -\frac{\hbar^2}{2m} \int_{-\infty}^{+\infty} \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx$$

而 $\Psi(x, t)$ 对 x 求偏导数得:

$$\begin{aligned}
 \frac{\partial \Psi}{\partial x} &= \left[-\frac{2a}{\gamma^2} \left(x - \frac{hlt}{m} \right) + il \right] \Psi \\
 \frac{\partial^2 \Psi}{\partial x^2} &= -\frac{2a}{\gamma^2} \Psi + \left[-\frac{2a}{\gamma^2} \left(x - \frac{hlt}{m} \right) + il \right]^2 \Psi = (Ax^2 + Bx + C) \Psi
 \end{aligned}$$

其中:

$$A = \frac{4a^2}{\gamma^4}, \quad B = -\frac{4ial}{\gamma^4}, \quad C = -\frac{2a\gamma^2 + l^2}{\gamma^4}$$

则:

$$\begin{aligned}
 \langle H \rangle &= -\frac{\hbar^2}{2m} \int_{-\infty}^{+\infty} \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx \\
 &= -\frac{\hbar^2}{2m} \int_{-\infty}^{+\infty} (Ax^2 + Bx + C) |\Psi|^2 dx \\
 &= -\frac{\hbar^2}{2m} (A \langle x^2 \rangle + B \langle x \rangle + C) \\
 &= \frac{\hbar^2}{2m} (a + l^2)
 \end{aligned}$$

为了求得 $\langle H^2 \rangle$, 可以在动量表象下求得:

$$\langle H^2 \rangle = \frac{\langle p^4 \rangle}{4m^2} = \frac{1}{4m^2} \int_{-\infty}^{+\infty} p^4 |\Phi(p, t)|^2 dp$$

其中:

$$\Phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ipx}{\hbar}} \Psi(x, t) dx$$

因为自由粒子对于任意正整数 n 有 $[p^n, H] = 0$, $\langle p^n \rangle$ 与时间无关, 所以可以用 $\Phi(p, 0)$ 代替 $\Phi(p, t)$:

$$\langle H^2 \rangle = \frac{1}{4m^2} \int_{-\infty}^{+\infty} p^4 |\Phi(p, 0)|^2 dp, \quad \Phi(p, 0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ipx}{\hbar}} \Psi(x, 0) dx$$

注意 $t = 0$ 时 $\gamma = 1$, 于是:

$$\begin{aligned}
 \Phi(p, 0) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ipx}{\hbar}} \Psi(x, 0) dx \\
 &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \sqrt{\frac{2a}{\pi}} \exp\left[-ax^2 + ilx - \frac{ipx}{\hbar}\right] dx \\
 &= K \int_{-\infty}^{+\infty} \exp\left[-ax^2 + ix\left(l - \frac{p}{\hbar}\right)\right] dx, \quad K = \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2a}{\pi}}
 \end{aligned}$$

由:

$$\int_{-\infty}^{+\infty} \exp(-D^2 x^2 + Ex) dx = \int_{-\infty}^{+\infty} \exp\left[-D^2 \left(x - \frac{E}{2D^2}\right)^2 + \frac{E^2}{4D^2}\right] dx = \frac{\sqrt{\pi}}{D} \exp\left(\frac{E^2}{4D^2}\right)$$

得:

$$\begin{aligned}
 \Phi(p, 0) &= K \int_{-\infty}^{+\infty} \exp\left[-ax^2 + ix\left(l - \frac{p}{\hbar}\right)\right] dx \\
 &= K \sqrt{\frac{\pi}{a}} \exp\left[-\frac{1}{4a} \left(l - \frac{p}{\hbar}\right)^2\right] \\
 &= \frac{1}{\sqrt{2a\hbar}} \sqrt{\frac{2a}{\pi}} \exp\left[-\frac{1}{4a} \left(l - \frac{p}{\hbar}\right)^2\right]
 \end{aligned}$$

实际上同理也可以求出:

$$\Phi(p, t) = \frac{1}{\sqrt{2a\hbar}} \sqrt{\frac{2a}{\pi}} \exp\left[-\frac{\gamma^2}{4a} \left(l - \frac{p}{\hbar}\right)^2 - \frac{ihlt}{2m} \left(l - \frac{p}{\hbar}\right)\right]$$

注意到 $\gamma^2 = 1 + i\theta = 1 + i\theta(t)$, $t = 0$ 时 $\gamma^2 = 1$, 且 $\gamma^2 + (\gamma^2)^* = 2\operatorname{Re}(\gamma^2) = 2$, 于是有:

$$|\Phi(p, 0)|^2 = |\Phi(p, t)|^2 = \frac{1}{\hbar\sqrt{2a\pi}} \exp\left[-\frac{1}{2a}\left(l - \frac{p}{\hbar}\right)^2\right]$$

故 H^2 的平均值为:

$$\begin{aligned}\langle H^2 \rangle &= \frac{1}{4m^2} \int_{-\infty}^{+\infty} p^4 |\Phi(p, 0)|^2 dp \\ &= \frac{1}{4m^2} \cdot \frac{1}{\hbar\sqrt{2a\pi}} \int_{-\infty}^{+\infty} p^4 \exp\left[-\frac{1}{2a}\left(l - \frac{p}{\hbar}\right)^2\right] dp \\ &= \frac{1}{4m^2} \cdot \frac{1}{\hbar\sqrt{2a\pi}} \int_{-\infty}^{+\infty} \hbar^5 (z + l)^4 e^{-\frac{z^2}{2a}} dz, \quad z = \frac{p}{\hbar} - l \\ &= \frac{1}{4m^2} \cdot \frac{\hbar^4}{\sqrt{2a\pi}} \int_{-\infty}^{+\infty} (z^4 + 4z^3l + 6z^2l^2 + 4zl^3 + l^4) e^{-\frac{z^2}{2a}} dz \\ &= \frac{1}{4m^2} \cdot \frac{\hbar^4}{\sqrt{2a\pi}} \left[(2a)^{\frac{5}{2}} \cdot \frac{3\sqrt{\pi}}{4} + 6l^2(2a)^{\frac{3}{2}} \cdot \frac{\sqrt{\pi}}{2} + l^4\sqrt{2a} \cdot \sqrt{\pi} \right] \\ &= \frac{\hbar^4}{4m^2} (3a^2 + 6al^2 + l^4)\end{aligned}$$

其中使用了:

$$\int_{-\infty}^{+\infty} u^3 e^{-u^2} du = 0, \quad \int_{-\infty}^{+\infty} u^4 e^{-u^2} du = \frac{3\sqrt{\pi}}{4}$$

于是有:

$$\sigma_H^2 = \langle H^2 \rangle - \langle H \rangle^2 = \frac{\hbar^4}{2m^2} (a^2 + 2al^2)$$

则:

$$\sigma_H^2 \sigma_x^2 = \frac{\hbar^4}{2m^2} (a^2 + 2al^2) \cdot \left(\frac{1}{4a} + \frac{a\hbar^2 t^2}{m^2} \right) = \frac{\hbar^2}{4} \left(\frac{\hbar l}{m} \right)^2 \cdot \left(1 + \frac{a}{2l^2} \right) \left[1 + \left(\frac{2a\hbar t}{m} \right)^2 \right]$$

而:

$$\sigma_H^2 \sigma_x^2 \geq \frac{\hbar^2}{4} \left(\frac{\hbar l}{m} \right)^2 = \frac{\hbar^2}{4} \frac{d\langle x \rangle}{dt}$$

这符合能量——时间不确定度关系。

33.(a) 由于测量到了本征值 a_1 , 这说明系统在测量后立即坍缩到本征值 a_1 对应的本征态 ψ_1 。

33.(b) 由题意:

$$\begin{cases} \frac{1}{5}(3\phi_1 + 4\phi_2) = \psi_1 \\ \frac{1}{5}(4\phi_1 - 3\phi_2) = \psi_2 \end{cases}$$

解得:

$$\begin{cases} \phi_1 = \frac{1}{5}(3\psi_1 + 4\psi_2) \\ \phi_2 = \frac{1}{5}(4\psi_1 - 3\psi_2) \end{cases}$$

注意系统此时处于 ψ_1 态，所以测量到 b_1 的概率为：

$$\begin{aligned}
 P_{b_1} &= \langle \phi_1 | \psi_1 \rangle^2 \\
 &= \left\langle \frac{1}{5}(3\psi_1 + 4\psi_2) \middle| \psi_1 \right\rangle^2 \\
 &= \left(\frac{3}{5} \langle \psi_1 | \psi_1 \rangle + \frac{4}{5} \langle \psi_2 | \psi_1 \rangle \right)^2 \\
 &= \left(\frac{3}{5} \cdot 1 + \frac{4}{5} \cdot 0 \right)^2 \\
 &= \frac{9}{25}
 \end{aligned}$$

其中使用了本征函数的正交归一性：

$$\langle \psi_i | \psi_j \rangle = \delta_{ij}$$

同理，测量到 b_2 的概率为：

$$\begin{aligned}
 P_{b_2} &= \langle \phi_1 | \psi_2 \rangle^2 \\
 &= \left\langle \frac{1}{5}(3\psi_1 + 4\psi_2) \middle| \psi_2 \right\rangle^2 \\
 &= \left(\frac{3}{5} \langle \psi_1 | \psi_2 \rangle + \frac{4}{5} \langle \psi_2 | \psi_2 \rangle \right)^2 \\
 &= \left(\frac{3}{5} \cdot 0 + \frac{4}{5} \cdot 1 \right)^2 \\
 &= \frac{16}{25}
 \end{aligned}$$

33.(c) 测量到 b_1 的概率为 $\frac{9}{25}$ ，则此时粒子的状态为 ϕ_1 ，测量到 a_1 的概率为：

$$\begin{aligned}
 P_{\phi_1 \Rightarrow a_1} &= \langle \psi_1 | \phi_1 \rangle^2 \\
 &= \left\langle \frac{1}{5}(3\phi_1 + 4\phi_2) \middle| \phi_1 \right\rangle^2 \\
 &= \left(\frac{3}{5} \langle \phi_1 | \phi_1 \rangle + \frac{4}{5} \langle \phi_2 | \phi_1 \rangle \right)^2 \\
 &= \left(\frac{3}{5} \cdot 1 + \frac{4}{5} \cdot 0 \right)^2 \\
 &= \frac{9}{25}
 \end{aligned}$$

测量到 b_1 的概率为 $\frac{16}{25}$ ，则此时粒子的状态为 ϕ_2 ，测量到 a_1 的概率为：

$$\begin{aligned}
 P_{\phi_2 \Rightarrow a_1} &= \langle \psi_2 | \phi_1 \rangle^2 \\
 &= \left\langle \frac{1}{5}(4\phi_1 - 3\phi_2) \middle| \phi_1 \right\rangle^2
 \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{4}{5} \langle \phi_1 | \phi_1 \rangle - \frac{3}{5} \langle \phi_2 | \phi_1 \rangle \right)^2 \\
&= \left(\frac{4}{5} \cdot 1 - \frac{3}{5} \cdot 0 \right)^2 \\
&= \frac{16}{25}
\end{aligned}$$

所以测量到 a_1 的总概率为:

$$\frac{9}{25} \cdot \frac{9}{25} + \frac{16}{25} \cdot \frac{16}{25} = \frac{337}{625} = 53.92\%$$

37. 由

$$\frac{d}{dt} \langle Q \rangle = \frac{i}{\hbar} \langle [H, Q] \rangle + \left\langle \frac{\partial Q}{\partial t} \right\rangle$$

得:

$$\frac{d}{dt} \langle xp \rangle = \frac{i}{\hbar} \langle [H, xp] \rangle + \left\langle \frac{\partial p}{\partial t} \right\rangle$$

因为 p 不含时, $\frac{\partial p}{\partial t} = 0$, 又 $[H, xp] = [H, x]p + x[H, p]$, 故有:

$$\frac{d}{dt} \langle xp \rangle = \frac{i}{\hbar} \langle [H, x]p + x[H, p] \rangle$$

分别计算 $[H, x]$ 与 $[H, p]$ 如下:

$$\begin{aligned}
[H, x]\Psi &= \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) x\Psi - x \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) \Psi \\
&= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (x\Psi) + x \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi \\
&= -\frac{\hbar^2}{2m} \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} (x\Psi) \right] + x \frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} \\
&= -\frac{\hbar^2}{2m} \frac{\partial}{\partial x} \left(x \frac{\partial \Psi}{\partial x} + \Psi \right) + x \frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} \\
&= -\frac{\hbar^2}{2m} \left(x \frac{\partial^2 \Psi}{\partial x^2} + 2 \frac{\partial \Psi}{\partial x} \right) + x \frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} \\
&= -\frac{\hbar^2}{m} \frac{\partial \Psi}{\partial x}
\end{aligned}$$

即:

$$[H, x] = -\frac{\hbar^2}{m} \frac{\partial}{\partial x} = -\frac{i\hbar p}{m}$$

又:

$$\begin{aligned}
[H, p]\Psi &= \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi - \left(-i\hbar \frac{\partial}{\partial x} \right) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) \Psi \\
&= \frac{i\hbar^3}{2m} \frac{\partial^3 \Psi}{\partial x^3} - i\hbar V \frac{\partial \Psi}{\partial x} - \frac{i\hbar^3}{2m} \frac{\partial^3 \Psi}{\partial x^3} + i\hbar \frac{\partial(V\Psi)}{\partial x} \\
&= i\hbar \frac{\partial(V\Psi)}{\partial x} - i\hbar V \frac{\partial \Psi}{\partial x}
\end{aligned}$$

$$\begin{aligned}
&= i\hbar V \frac{\partial \Psi}{\partial x} + i\hbar \Psi \frac{\partial V}{\partial x} - i\hbar V \frac{\partial \Psi}{\partial x} \\
&= i\hbar \frac{\partial V}{\partial x} \Psi
\end{aligned}$$

即:

$$[H, p] = i\hbar \frac{\partial V}{\partial x}$$

代入得:

$$\begin{aligned}
\frac{d}{dt} \langle xp \rangle &= \frac{i}{\hbar} \langle [H, x]p + x[H, p] \rangle \\
&= \frac{i}{\hbar} \left\langle -\frac{i\hbar p}{m} + i\hbar \frac{\partial V}{\partial x} \right\rangle \\
&= 2 \left\langle \frac{p^2}{2m} \right\rangle - \left\langle x \frac{\partial V}{\partial x} \right\rangle \\
&= 2 \langle T \rangle - \left\langle x \frac{\partial V}{\partial x} \right\rangle
\end{aligned}$$

在定态下, 一切不显含时间的力学量的平均值和概率分布都不随时间改变, 因此 $\frac{d}{dt} \langle xp \rangle = 0$, 此时有位力定理:

$$2 \langle T \rangle = \left\langle x \frac{\partial V}{\partial x} \right\rangle$$

对于谐振子:

$$V = \frac{1}{2} m \omega^2 x^2$$

则有:

$$x \frac{dV}{dx} = x \cdot m \omega^2 x = m \omega^2 x^2 = 2V$$

由于 $2 \langle T \rangle = \left\langle x \frac{\partial V}{\partial x} \right\rangle$, 得 $2 \langle T \rangle = \langle 2V \rangle$, 即:

$$\langle T \rangle = \langle V \rangle$$

对于谐振子的定态:

$$\langle T \rangle = \langle V \rangle = \frac{1}{2} \left(n + \frac{1}{2} \right) \hbar \omega$$

49.(a) 对于自由粒子, $V(x) = 0$, 在坐标表象下, Schrödinger 方程为:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2}$$

在动量表象下, 用 $\Phi(p, t)$ 代替 $\Psi(x, t)$, 用 p 代替 $-i\hbar \frac{\partial}{\partial x}$, Schrödinger 方程变为:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{p^2}{2m} \Psi$$

解得:

$$\Phi(p, t) = \Phi(p, 0) e^{-\frac{ip^2 t}{2m\hbar}}$$

49.(b) 已知波函数:

$$\Psi(x, t) = \sqrt[4]{\frac{2a}{\pi}} \frac{1}{\gamma} \exp \left[-\frac{a}{\gamma^2} \left(x - \frac{hlt}{m} \right)^2 + il \left(x - \frac{hlt}{2m} \right) \right], \quad \gamma = \sqrt{1 + \frac{2ai\hbar t}{m}}$$

记 $\theta = \frac{2a\hbar t}{m}$, 则 $\gamma = \sqrt{1 + i\theta}$, 注意 $t = 0$ 时 $\gamma = 1$, 于是:

$$\begin{aligned} \Phi(p, 0) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ipx}{\hbar}} \Psi(x, 0) dx \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \sqrt[4]{\frac{2a}{\pi}} \exp \left[-ax^2 + ilx - \frac{ipx}{\hbar} \right] dx \\ &= K \int_{-\infty}^{+\infty} \exp \left[-ax^2 + ix \left(l - \frac{p}{\hbar} \right) \right] dx, \quad K = \frac{1}{\sqrt{2\pi\hbar}} \sqrt[4]{\frac{2a}{\pi}} \end{aligned}$$

由:

$$\int_{-\infty}^{+\infty} \exp(-D^2 x^2 + Ex) dx = \int_{-\infty}^{+\infty} \exp \left[-D^2 \left(x - \frac{E}{2D^2} \right)^2 + \frac{E^2}{4D^2} \right] dx = \frac{\sqrt{\pi}}{D} \exp \left(\frac{E^2}{4D^2} \right)$$

得:

$$\begin{aligned} \Phi(p, 0) &= K \int_{-\infty}^{+\infty} \exp \left[-ax^2 + ix \left(l - \frac{p}{\hbar} \right) \right] dx \\ &= K \sqrt{\frac{\pi}{a}} \exp \left[-\frac{1}{4a} \left(l - \frac{p}{\hbar} \right)^2 \right] \\ &= \frac{1}{\sqrt{2a\hbar}} \sqrt[4]{\frac{2a}{\pi}} \exp \left[-\frac{1}{4a} \left(l - \frac{p}{\hbar} \right)^2 \right] \end{aligned}$$

同理也可以求出:

$$\begin{aligned} \Phi(p, t) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ipx}{\hbar}} \Psi(x, t) dx \\ &= K \int_{-\infty}^{+\infty} \exp \left[-\frac{a}{\gamma^2} \left(x - \frac{hlt}{m} \right)^2 + il \left(x - \frac{hlt}{2m} \right) - \frac{ipx}{\hbar} \right] dx \\ &= K \exp \left[-\frac{a}{\gamma^2} \left(\frac{hlt}{m} \right)^2 + il \frac{hlt}{2m} \right] \int_{-\infty}^{+\infty} \exp \left[-ax^2 + ix \left(l - \frac{p}{\hbar} \right) \right] dx \\ &= K \frac{\gamma \sqrt{\pi}}{\sqrt{a}} \exp \left[-\frac{a}{\gamma^2} \left(\frac{hlt}{m} \right)^2 + il \frac{hlt}{2m} \right] \exp \left\{ \frac{\gamma^2}{4a} \left[i \left(l - \frac{p}{\hbar} \right) - \frac{2a}{\gamma^2} \left(\frac{hlt}{m} \right) \right]^2 \right\} \\ &= \frac{1}{\sqrt{2a\hbar}} \sqrt[4]{\frac{2a}{\pi}} \exp \left[-\frac{\gamma^2}{4a} \left(l - \frac{p}{\hbar} \right)^2 - \frac{ihlt}{2m} \left(l - \frac{p}{\hbar} \right) \right] \end{aligned}$$

注意到 $\gamma^2 = 1 + i\theta = 1 + i\theta(t)$, $t = 0$ 时 $\gamma^2 = 1$, 且 $\gamma^2 + (\gamma^2)^* = 2\text{Re}(\gamma^2) = 2$, 于是有:

$$|\Phi(p, 0)|^2 = |\Phi(p, t)|^2 = \frac{1}{\hbar \sqrt{2a\pi}} \exp \left[-\frac{1}{2a} \left(l - \frac{p}{\hbar} \right)^2 \right]$$

可见 $|\Phi(p, t)|^2$ 与时间无关。

49.(c) p 的平均值为:

$$\begin{aligned}
 \langle p \rangle &= \int_{-\infty}^{+\infty} \langle \Phi | p | \Phi \rangle dp \\
 &= \int_{-\infty}^{+\infty} p |\Phi(p, 0)|^2 dp \\
 &= \frac{1}{\hbar \sqrt{2a\pi}} \int_{-\infty}^{+\infty} p \exp \left[-\frac{1}{2a} \left(l - \frac{p}{\hbar} \right)^2 \right] dp \\
 &= \frac{1}{\hbar \sqrt{2a\pi}} \int_{-\infty}^{+\infty} p \exp \left[-\frac{1}{2a} \left(\frac{p}{\hbar} - l \right)^2 \right] dp \\
 &= \frac{1}{\hbar \sqrt{2a\pi}} \int_{-\infty}^{+\infty} \hbar^2 (u + l) e^{-\frac{1}{2a} u^2} du, \quad u = \frac{p}{\hbar} - l \\
 &= 0 + \frac{\hbar l}{\sqrt{2a\pi}} \cdot \sqrt{2a\pi} \\
 &= \hbar l
 \end{aligned}$$

同理, p^2 的平均值为:

$$\begin{aligned}
 \langle p^2 \rangle &= \frac{1}{\hbar \sqrt{2a\pi}} \int_{-\infty}^{+\infty} p^2 \exp \left[-\frac{1}{2a} \left(\frac{p}{\hbar} - l \right)^2 \right] dp \\
 &= \frac{1}{\hbar \sqrt{2a\pi}} \int_{-\infty}^{+\infty} \hbar^3 (u + l)^2 e^{-\frac{1}{2a} u^2} du, \quad u = \frac{p}{\hbar} - l \\
 &= \frac{\hbar^2}{\sqrt{2a\pi}} \cdot \frac{a}{2} \sqrt{2a\pi} + 0 + \frac{\hbar^2}{\sqrt{2a\pi}} \cdot l^2 \sqrt{2a\pi} \\
 &= \hbar^2 (a + l^2)
 \end{aligned}$$

49.(d) Hamilton 量的平均值为:

$$\begin{aligned}
 \langle H \rangle &= \frac{\langle p^2 \rangle}{2m} \\
 &= \frac{\hbar^2}{2m} (a + l^2) \\
 &= \frac{\langle p \rangle^2}{2m} + \frac{a\hbar^2}{2m}
 \end{aligned}$$

而对于静态高斯波包的波函数:

$$\Psi_0(x, t) = \sqrt{\frac{2a}{\pi}} \frac{1}{\gamma'} e^{-\frac{a}{\gamma'^2} x^2}, \quad \gamma' = \sqrt{1 + \frac{2ai\hbar t}{m}}$$

Hamilton 量的平均值为 $\langle H \rangle_0 = \frac{a\hbar^2}{2m}$, 故:

$$\langle H \rangle = \frac{\langle p \rangle^2}{2m} + \langle H \rangle_0$$

第二章

2. 对于:

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\psi$$

如果 $E < \min V(x)$, 则 $\frac{2m}{\hbar^2}[V(x) - E] > 0$, ψ 与 $\psi''(x) = \frac{d^2\psi}{dx^2}$ 同号。此时分两种情况:

① $\psi(x)$ 有零点 x_0 :

则存在实数 $\delta > 0$, 使得在 $x \in (x_0, x_0 + \delta)$ 处, $\psi(x)$ 恒大于零或恒小于零。

如果在 $x \in (x_0, x_0 + \delta)$ 处, $\psi(x)$ 恒大于零, 由 $\psi(x_0 + \delta) = \psi(x_0) + \delta\psi'(x_0) + o(\delta)$ 得 $\psi'(x_0) > 0$ 。

注意到此时 $\psi''(x)$ 也大于零, 则 $\psi(x)$ 在 $(x_0, +\infty)$ 上单调递增且 $\psi'(x)$ 越来越大。

这使得 $\lim_{x \rightarrow +\infty} \psi(x) = +\infty$, $\psi(x)$ 不可归一化。

同理, 如果在 $x \in (x_0, x_0 + \delta)$ 处, $\psi(x)$ 恒小于零, 则 $\psi'(x) < 0$, $\psi''(x) < 0$, $\lim_{x \rightarrow +\infty} \psi(x) = -\infty$, $\psi(x)$ 也不可归一化。

② $\psi(x)$ 没有零点:

则 $\psi(x)$ 在 $(-\infty, +\infty)$ 上恒大于零或恒小于零, $\psi''(x)$ 在 $(-\infty, +\infty)$ 上也恒大于零或恒小于零。

于是 $\psi(x)$ 恒大于零且 $\psi'(x)$ 单调递增, 这使得 $\lim_{x \rightarrow +\infty} \psi(x) = +\infty$, $\psi(x)$ 不可归一化。

同理, $\psi(x)$ 恒小于零且 $\psi'(x)$ 单调递减, 这使得 $\lim_{x \rightarrow +\infty} \psi(x) = -\infty$, $\psi(x)$ 不可归一化。

综上, 如果 $E < \min V(x)$, 则 ψ 与 $\psi''(x)$ 同号, 这样的 $\psi(x)$ 不可归一化, 故 $E \geq \min V(x)$ 。

4. 由波函数:

$$\Psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t}, \quad n \in \mathbb{N}^*$$

得:

$$\begin{aligned} \langle x \rangle &= \int_0^a \langle \Psi_n | x | \Psi_n \rangle dx \\ &= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot x \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{i\omega t} dx \\ &= \frac{2a}{(n\pi)^2} \int_0^{n\pi} u \sin^2 u du, \quad u = \frac{n\pi x}{a} \\ &= \frac{2a}{(n\pi)^2} \left(\frac{u^2}{4} - \frac{u}{4} \sin 2u - \frac{1}{8} \cos 2u \right) \Big|_0^{n\pi} \\ &= \frac{2a}{(n\pi)^2} \cdot \frac{(n\pi)^2}{4} \\ &= \frac{a}{2} \end{aligned}$$

$$\begin{aligned}
\langle x^2 \rangle &= \int_0^a \langle \Psi_n | x^2 | \Psi_n \rangle dx \\
&= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot x^2 \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{i\omega t} dx \\
&= \frac{2a^2}{(n\pi)^3} \int_0^{n\pi} u \sin^2 u du, \quad u = \frac{n\pi x}{a} \\
&= \frac{2a^2}{(n\pi)^3} \left(\frac{u^3}{6} - \frac{2u^2 - 1}{8} \sin 2u - \frac{u}{4} \cos 2u \right) \Big|_0^{n\pi} \\
&= \frac{2a^2}{(n\pi)^3} \cdot \left[\frac{(n\pi)^3}{6} - \frac{n\pi}{4} \right] \\
&= \frac{a^2}{3} - \frac{1}{2n^2\pi^2}, \quad n \in \mathbb{N}^*
\end{aligned}$$

$$\begin{aligned}
\langle p \rangle &= \int_0^a \langle \Psi_n | p | \Psi_n \rangle dx \\
&= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot (-i\hbar) \frac{\partial}{\partial x} \left(\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{i\omega t} \right) dx \\
&= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot (-i\hbar) \sqrt{\frac{2}{a}} \left(\frac{n\pi}{a}\right) \cos\left(\frac{n\pi x}{a}\right) e^{i\omega t} dx \\
&= -\frac{2i\hbar}{a} \int_0^{n\pi} \sin u \cos u du, \quad u = \frac{n\pi x}{a} \\
&= -\frac{2i\hbar}{a} \left(-\frac{1}{2} \cos^2 u \right) \Big|_0^{n\pi} \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\langle p^2 \rangle &= \int_0^a \langle \Psi_n | p^2 | \Psi_n \rangle dx \\
&= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot (-\hbar^2) \frac{\partial^2}{\partial x^2} \left(\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{i\omega t} \right) dx \\
&= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\omega t} \cdot \hbar^2 \sqrt{\frac{2}{a}} \left(\frac{n\pi}{a}\right)^2 \sin\left(\frac{n\pi x}{a}\right) e^{i\omega t} dx \\
&= \frac{2n\pi\hbar^2}{a^2} \int_0^{n\pi} \sin^2 u du, \quad u = \frac{n\pi x}{a} \\
&= \frac{2n\pi\hbar^2}{a^2} \left(\frac{u}{2} - \frac{1}{4} \sin 2u \right) \Big|_0^{n\pi} \\
&= \frac{n^2\pi^2\hbar^2}{a^2}, \quad n \in \mathbb{N}^*
\end{aligned}$$

于是有：

$$\sigma_x = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a}{2} \sqrt{\frac{1}{3} - \frac{2}{n^2\pi^2}}, \quad n \in \mathbb{N}^*$$

$$\sigma_p = \langle p^2 \rangle - \langle p \rangle^2 = \frac{n^2 \pi^2 \hbar^2}{a^2}, \quad n \in \mathbb{N}^*$$

则:

$$\sigma_x \sigma_p = \frac{a}{2} \sqrt{\frac{1}{3} - \frac{2}{n^2 \pi^2}} \cdot \frac{n^2 \pi^2 \hbar^2}{a^2} = \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}, \quad n \in \mathbb{N}^*$$

$n = 1$ 时 $\sigma_x \sigma_p$ 最小:

$$(\sigma_x \sigma_p)_{\min} = \frac{\hbar}{2} \sqrt{\frac{\pi^2}{3} - 2} = 1.136 \frac{\hbar}{2} > \frac{\hbar}{2}$$